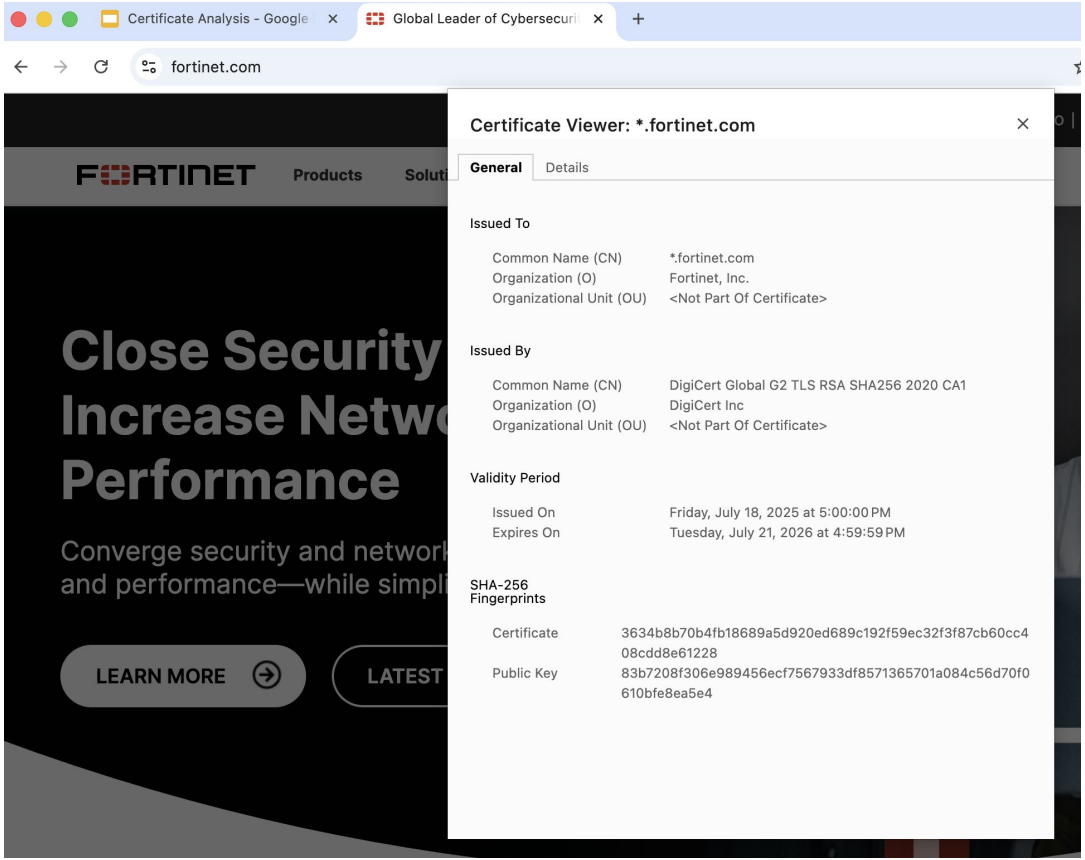
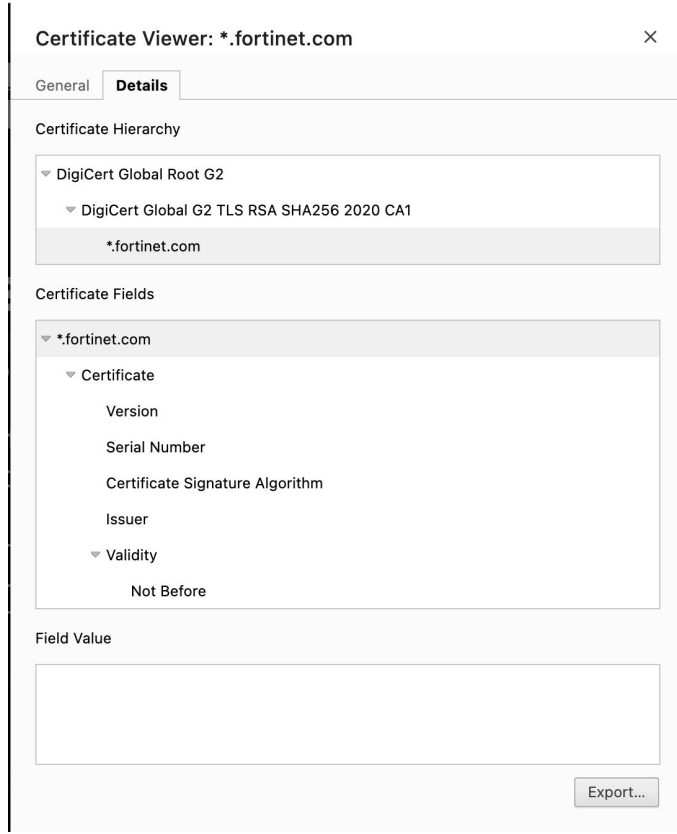
Certificate analysis: fortinet.com

Shinyoung You


Certificate Analysis - Google
Global Leader of Cybersecuri
fortinet.com
FORTINET
Products
Soluti
Close Security
Increase Netwo
Performance
Converge security and networ
and performance—while simpli
LEARN MORE
LATEST
Certificate Viewer: *.fortinet.com
General
Details
Issued To
Common Name (CN) *.fortinet.com
Organization (O) Fortinet, Inc.
Organizational Unit (OU) <Not Part Of Certificate>
Issued By
Common Name (CN) DigiCert Global G2 TLS RSA SHA256 2020 CA1
Organization (O) DigiCert Inc
Organizational Unit (OU) <Not Part Of Certificate>
Validity Period
Issued On Friday, July 18, 2025 at 5:00:00 PM
Expires On Tuesday, July 21, 2026 at 4:59:59 PM
SHA-256 Fingerprints
Certificate 3634b8b70b4fb18689a5d920ed689c192f59ec32f3f87cb60cc408cdd8e61228
Public Key 83b7208f306e989456ecf7567933df8571365701a084c56d70f0610bfe8ea5e4


Certificate Viewer: *.fortinet.com
General
Details
Certificate Hierarchy
DigiCert Global Root G2
DigiCert Global G2 TLS RSA SHA256 2020 CA1
*.fortinet.com
Certificate Fields
*.fortinet.com
Certificate
Version
Serial Number
Certificate Signature Algorithm
Issuer
Validity
Not Before
Field Value
Export...

Certificate Chain

DigiCert Global Root G2 (Root CA — trusted by browsers/OS)
    └── DigiCert Global G2 TLS RSA SHA256 2020 CA1 (Intermediate CA)
        └── *.fortinet.com (End-entity / Leaf Certificate)

Root CA stays offline for security, it only signed the intermediate.
Intermediate CA is what actually signed the Fortinet leaf cert

Leaf Certificate: *.fortinet.com

- Common Name: *.fortinet.com (*: wildcard, covers all first-level subdomains)
  - e.g. docs.fortinet.com, support.fortinet.com [v] a.b.fortinet.com [x]
- Organization: Fortinet, Inc.
- Issuer: DigiCert Global G2 TLS RSA SHA256 2020 CA1
- Issued: July 18, 2025
- Expires: July 21, 2026
- Validity Duration: ~1 year (standard, aligns with CA/Browser Forum limits)
- Cert SHA-256: 3634b8b7...08cdd8e61228
- Public Key SHA-256: 83b7208f...610bfe8ea5e4

Intermediate CA: DigiCert Global G2 TLS RSA SHA256 2020 CA1

Parsed from the exported.pem file:

- Serial Number: 0CF5BD062B5602F47AB8502C23CCF066
- Issued By: DigiCert Global Root G2
- Valid From: March 30, 2021
- Valid Until: March 29, 2031 (10-year lifespan, typical for intermediates)
- Public Key: RSA 2048-bit
- Key Usage: Digital Signature, Certificate Sign, CRL Sign
- Extended Key Usage: TLS Web Server + Client Authentication
- Path Length Constraint:
    - pathlen:0, meaning cannot issue further sub-CAs
- OCSP: http://ocsp.digicert.com
- CRL: http://crl3.digicert.com/DigiCertGlobalRootG2.crl

Algorithm: RSA with SHA-256

RSA

- Asymmetric encryption (encrypt -> decrypt)
- Uses a public + private key pair
  - Signer (DigiCert) signs the digest with their own private key
  - Receiver verifies using DigiCert's public key (RSA 2048-bit)

SHA-256

- Hash function: one-way, not reversible
- Takes input certificate data and produces a fixed 256-bit digest

Revocation check: OCSP and CRL

OCSP — Online Certificate Status Protocol

Browser → asks ocsp.digicert.com → "is this cert still valid?"

DigiCert → responds → "good" / "revoked" / "unknown"

- Real-time, per-certificate check
- Fast: just a small request/response
- Privacy concern: DigiCert can see which sites you're visiting

CRL — Certificate Revocation List

Browser → downloads the full list from DigiCert → checks if cert's serial number is in the list

- A periodically published file containing all revoked cert serial numbers
- Slower and heavier (the list can be large)
- But no real-time callout needed once downloaded, can be cached